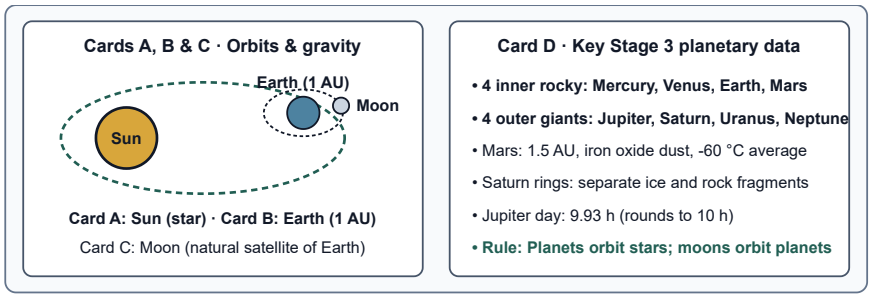
Reference sheets

Reference A · Class Solar System Source Cards (Cards A, B, C, and D)

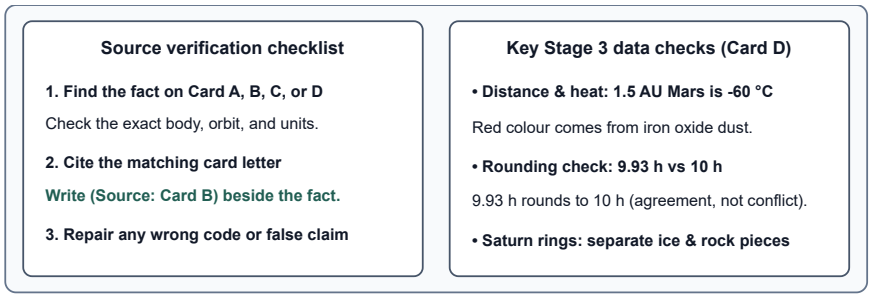


Class Source Cards A, B, C, and D showing the Sun's gravity, Earth's 1 AU orbit, the Moon's satellite orbit, and Key Stage 3 planetary data.

Reference only. Do not write on this sheet. Read the source card first, then cite its exact code beside your fact.

Source card code	Body and classification	Verified science facts on this card	Key Stage 3 mechanism or measurement
Card A · Star card	The Sun (central star)	The Sun is a hot star at the centre of our Solar System that emits its own light	The Sun's non-contact gravitational pull holds planets in curved orbits around it
Card B · Planet card	Earth (inner rocky planet)	Earth is a nearly spherical rocky planet that orbits the Sun once a year	Earth orbits at an average distance of 1 astronomical unit (1 AU, about 150 million km)
Card C · Moon card	The Moon (natural satellite)	The Moon is a rocky natural satellite that orbits Earth in about a month and reflects sunlight	Earth's gravity holds the Moon in orbit around Earth rather than directly around the Sun
Card D · Solar System data	4 inner rocky planets & 4 outer giants	Inner rocky: Mercury, Venus, Earth, Mars. Outer giants: Jupiter, Saturn, Uranus, Neptune	Mars (1.5 AU, -60 °C, iron oxide dust); Saturn rings (ice and rock fragments); Jupiter day (9.93 h, rounds to 10 h)

Reference B · Key Stage 3 source verification and claim repair guide



Key Stage 3 source verification guide: 1. Find the fact on Cards A to D, 2. Cite the card code, 3. Repair wrong citations or false claims.

Research check	Faulty or incomplete note	How to verify and repair using Cards A, B, C, and D
Missing source code	'Earth orbits the Sun at 1 AU.' (no source named)	Add the matching card code so any reader can check it: 'Earth orbits the Sun at 1 AU (Source: Card B).'
Wrong source code	'The Moon orbits Earth (Source: Card A).'	Card A is about the Sun. Replace the code with Card C: 'The Moon orbits Earth (Source: Card C).'
False planet claim	'The Moon is a planet because it is round.'	Cards B, C, and D show that planets orbit a star (the Sun), whereas the Moon orbits a planet (Earth) as a natural satellite.
False temperature or ring claim	'Mars is red so it is hot; Saturn has a solid ring.'	Card D shows Mars is red from iron oxide dust and cold (-60 °C at 1.5 AU), and Saturn's rings are separate ice and rock pieces.
Rounded data comparison	Table 1 gives Jupiter's day as 9.93 h; Table 2 gives 10 h.	Card D confirms both agree after rounding: 9.93 hours rounds to 10 hours to the nearest whole hour.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. From the arrival picture cue, state two visual differences between the central body on Card A and the orbiting bodies on Cards B and C.

Help: Compare the light rays on Card A with the dashed orbit paths on Cards B and C.

2. What non-contact force holds planets in orbit around the Sun (Card B) and moons in orbit around planets (Card C)?

Help: Recall the Key Stage 3 force that acts between masses across space.

3. Earth orbits at 1 astronomical unit (1 AU) from the Sun, while Mars orbits at 1.5 AU. How does greater orbital distance affect solar heating?

Help: Link distance from the Sun in AU to surface temperature.

4. Why do classroom diagrams of Cards A, B, and C include the note 'Not to scale'?

Help: Compare the real size and 1 AU distance of the Sun and Earth with a page drawing.

You do tasks

1. Use Reference A and Reference B to compare the four inner rocky planets with the four outer giant planets, explaining how gravity and orbital distance (1 AU vs 1.5 AU) shape the Solar System. **Staff I ☐ S ☐ C ☐**

Help: Link the Sun's gravity (Card A) and orbital distance in AU (Earth at 1 AU on Card B; Mars at 1.5 AU and -60 °C on Card D) to planet composition and temperature.

2. Two astronomy data tables on Card D list Jupiter's rotation period (length of day) as '9.93 hours' in Table 1 and 'about 10 hours' in Table 2. Explain whether these two sources contradict each other or agree after rounding. **Staff I ☐ S ☐ C ☐**

Help: Compare 9.93 hours rounded to the nearest whole hour with 10 hours on Reference B.

3. Evaluate and repair a flawed briefing slide that says: 'The Moon is a hot planet that orbits the Sun at 1 AU, and Mars is red because it is boiling.' Cite the exact cards that disprove each error. **Staff I ☐ S ☐ C ☐**

Help: Break the flawed briefing into its Moon natural satellite claim (Cards B and C) and its Mars iron oxide and -60 °C at 1.5 AU claim (Card D).

Optional participation-loop prompt: Share one verified fact and source code with an adult, hear one feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. How do Card B, Card C, and Card D distinguish an inner rocky planet from a natural satellite and an outer giant planet?

2. Use Card D to explain why Mars (-60 °C at 1.5 AU) is colder than Earth (1 AU) even though Mars looks reddish.

3. Why do data tables listing Jupiter's day as 9.93 hours and 10 hours agree rather than contradict each other?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt